Alerts systems against crypto exploits

Market estimation and growth drivers

Market

$131.6M

- B2B
  - Protocols – $123.7M
  - Building blocks – $1.3M
  - Institutional – $10.1M
- B2C – $0.2M
  - *Very slow adoption*
  - *Very tough learning curve*

Protocols

$123.7M

1 | Protocols: own risks $120M

Estimation formula: total = auditing + actual bug bounties + potential bug bounties = $60M + $23M + $37M = $120M

Own risks: will be hacked → Current solution: audits and bug bounties

NEWS NEWSLETTER PODCASTS **Blockworks**

327.92 6.06% ADA 0.40 6.27% SOL 24.97 7.36% AVAX 22.23 16.43% UNI 6.43 -2.60% AAVE 90.48 11.48% COMP 55.16 6.98% S&P 500 4173.11 1.33% GOLD 1939

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DeFi Platform Exploited for $14.5M Despite Security Audits

Despite having had several firms conduct security audits of its smart contract code, Team Finance has been exploited for millions

1. Smart contracts **auditing** market $60M (annual)

2. Smart contracts **bug bounties** market $23M* (annual)

Potential bug bounties = total hacks (2022) **x** *bug bounty reward = 3.7B x 1% = 37M*

3. Total hacks in 2022: $3.7B → 1% as a bug bounty reward → 37M MARKET

Sources: blockworks, securityweek, Alchemy

**Assuming 14 large auditing companies in the market + the same auditing volume provide small and medium auditing companies*

1 | Protocols: integration risk $3.7M

Integration risk: **affiliated protocols** will be hacked → No control over other protocols' security!

Estimation formula: total hacks (2022) ***x*** *risk premium (as a % of potential hack) = $3.7B* ***x*** *0.1% = $3.7M*

*Ex: BNB Chain Hack, Oct 22**

The hack cost the company $570 million and happened due to an unexpected problem on the network that allowed the attacker to create 2 million BNB tokens out of nothing.
The root cause of the hack was identified as <u>a bug in the bridge</u>'s smart contract.

Sources: 1. blaze.tech

Key growth drivers: protocols

Progressing DeFi complexity

The higher is the system's complexity, the more fragile it is.

- Growth of cross-chain and bridging protocols as a way to meet interoperability needs and expectations;

A tiny share of TradFi ($100T+) is represented in DeFi: Consumer Banking, Capital Markets, and Insurance are waiting to be disrupted by blockchain players.

The wealthier is the industry the more is the incentive for malicious actors to hack it.

- DeFi market was estimated at $13.6B* in 2022, growing 45%** Y-o-Y;
- Transactions volume growth 37x*** 2015-o-2021

Following similar trends … DeFi market will achieve $500B** by 2028**

Sources: 1. Grandviewresearch 2. bccresearch 3. Cointelegraph 4. emergenresearch

Institutional

$10.1M

More than $1.14T traded*
by Institutional Investors annually

Sources: Coinbase

2 | Institutional

- Hedge Funds
- Some Venture Funds

(invests in tokens or hold some AUM in cryptocurrency)

>

Want to get ahead of

Hacks
Market manipulations

Both for Hedge Funds and Venture Funds, assets safety is sometimes about making exit at the right moment.

Use cases

		Hedge Funds	Venture Funds
exit before hack happens	Safe assets that are stored in cryptocurrency	+	+
	Safe assets that are invested in DeFi protocols, staking, etc.	+	–
be aware of huge market manipulations	Safe assets while market manipulations	+	–
	Get market insights: get earnings	+	–

Institutional: Hedge Funds $4.1M

Estimation formula: total fund's assets in crypto ***x*** *share of assets management fee = $4.1B* ***x*** *0.1% = $4.1M*

Hedge Funds' investments in crypto assets are growing:

- **38%*** of traditionals hedge funds invested in digital assets in 2022 (vs 21% in 2021);
- The median crypto AuM **tripled** 2022-o-2021;

Hedge Funds use software to get market insights and manage risks.

- The Hedge Funds Software Market $1.1B****

Hedge Funds suffer from crypto scam and hacks

- $4.1B crypto AUM of Hedge Funds

Hedge Funds' losses in 2022 (examples)

FTX

*Between 25% and 40% of cryptocurrency-focused hedge funds had some level of direct exposure to FTX or the exchange's native token, FTT.***

Terra

*In April 2022 (before the collapse) 27% of crypto hedge funds used TerraUSD (UST) as part of their trading activities.****

Sources: 1.PwC 2.Blockworks 3.tradersmagazine 4.businessresearchinsights

Institutional: Venture Funds $6M

Estimation formula: total fund's assets in crypto ***x*** *share of assets management fee = $6B* ***x*** *0.1% = $6M*

Crypto Venture Funds

- Hold AUM in crypto (that was raised in crypto or is supposed to be invested in company's token).
- Invest in startups' tokens (together with equity or instead of equity).

Crypto Venture Funds suffer from crypto scam and hacks

- Multicoin Capital was unable to pull all of its crypto held on FTX before withdrawals were halted, saving just 24% of the funds held there*

VCs activity is stable despite stormy market conditions

- $4.1B $30B invested by VCs into 3000 crypto and blockchain startups in 2022**.
- At least $36.1B raised by crypto Venture Funds in 2022***.

Sources: 1. Blockworks 2.galaxy 3. cointelegraph

Tier 1 Financial Institutions adopt crypto

Markets

European Investment Bank Issues $121M Digital Notes Using Ethereum

Goldman Sachs, Banco Santander SA and Societe Generale AG served as joint managers.

By Tanzeel Akhtar Apr 28, 2021 at 8:01 p.m. Updated Sep 14, 2021 at 3:48 p.m.

Bloomberg Middle East Edition

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Wealth
Crypto

Goldman Offers Its First Bitcoin-Backed Loan in Crypto Push

- Secured lending facility lent cash collateralized by Bitcoin
- Move comes as Wall Street warms up to digital assets

FORBES > MONEY > PERSONAL FINANCE

Goldman Sachs Co-Signs Bitcoin & Ethereum For High-Net Worth Clients

Sources: coindesk, forbes, bloomberg

Web2 corporations adopt crypto

Tech

Visa Proposes Automatic Payments Using Ethereum Layer 2 System StarkNet

Visa said self-custodial wallets could use a unique “account abstraction” method to set up automatic recurring payments on StarkNet as existing smart contracts do not currently support such steps.

By Shaurya Malwa Dec 20, 2022

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Meta to Let Users Mint and Sell Polygon-Powered NFTs on Instagram

Polygon Team Ecosystem
Nov 2, 2022

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Sarah Perez @sarahintampa / 5:00 PM GMT+2 • December 8, 2022 Comment

Sources: coindesk, polygon, TechCrunch

Building blocks

$1.3M

3 | Building blocks

How it works today (ethereum)

Searchers > Builders > Validator (consensus clients)

Tough competition for the block space.

New mempool insights contribute to their probability to win.

3 | Building blocks $1.3M

Estimation formula: total MEV extracted (annual) ***x*** *share of MEV searcher are ready to give up to get insights to get their tx included = $129M x 1% = $1.29M*

- $687M* of MEV extracted in total.
- $129M* extracted within 2022.
- 91%** of blocks are constructed using MEV-boost (special software by flashbots).

Growth drivers

- The Hedge However current MEV landscape includes mostly ethereum chain while MEV opportunities exist on each chain, DEX, bridge, oracle, etc.
- Decentralizing sequencers is one of core priorities for layers 2 within the next 2-3 years to be truly decentralized***

Sources: 1. MEV explorer 2. Mevboost.pics 3. twitter

B2C

$0.18M

4 | Regular crypto users

Estimation formula: total number of wallets **x** *share of paranoiacs* **x** *average amount of assets per wallet* **x** *risk premium (as a % of total assets held) =*
= 1,378k x 1% x $650 x 1% = $0.18M

Assume that at least the same amount of wallets are not active in a hold only mode. Hence, about 1,378k wallets in total.

Number of active wallets
ETH: 347k*
BTC: 815k****
BSC: 216k***

Assets volume (top 10 chains): $897B*****

Hack ratio**: 0.4%

Sources: 1. Dune 2. Assets_hacked/Total_assets 3. Dune 4. Bitinfocharts 5. coinmarketcap

Why now?

Blockchain market becomes more mature

- Regulation becomes clearer, governments collaborate with web3 giants (ex. Binance) to work on crypto initiatives together;

Conservative institutions (web2 enterprise) entering blockchain are much more risk-averse than crypto-native folks and use double-checking and risk-managing tools to mitigate the regular industry's risk level;

Tons of scam in 2022 made people more sceptical and careful about their investments, reputation, and new opportunities. Now they pay more attention to risk management and mitigation;

Right point in a market cycle: ahead of the new bull phase

- B2B deals are time-consuming: outreach, piloting, negotiations. Starting now we work ahead of the curve to be on time to meet the next market cycle.

CeFi

Opens more opportunities

Today: each protocol monitors fraud as it wants.

As the market becomes more mature: regulators will create fraud-monitoring standards (as for traditional and digital banking today) including pre-chain monitoring.

Spotter can cover the pre-chain monitoring stage and become a part of the compliance standard.

Disclaimer: this is an emerging market (compliance) inside of the emerging market (blockchain). However, we will keep an eye on what is happening there, as the volume of the CeFi market is 20x larger than the DeFi market and governments and financial institutions are actively working on blockchain industry transparent regulations.